Уравнения с малым параметром при производных

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In[∘]:=
```
ClearAll["Global`*"]
```

Условие задачи

In[∘]:=
```
Eq = μ^4 * D[y[x], {x, 4}] + ρ[x] * y[x] == 0
```

Out[∘]=

$$y[x] \times \rho[x] + \mu^4 y^{(4)}[x] == 0$$

Ищем решение в виде:

In[∘]:=
```
Y = Exp[Λ[x] / μ] * Sum[y[k][x] * μ^k, {k, 0, 0}]
```

Out[∘]=

$$e^{\frac{\Lambda[x]}{\mu}} y[0][x]$$

In[∘]:=
```
Eq1 = Simplify[Eq /. Table[D[y[x], {x, i}] → D[Y, {x, i}], {i, 0, 4}]] /. Join[{Exp[Λ[x] / μ] → 1},
    Table[D[Λ[x], {x, i}] → D[λ[x], {x, i - 1}], {i, 4}]];
EqArray = Thread[CoefficientList[Eq1⟦1⟧, μ] == 0];
```

In[∘]:=
```
TableForm[EqArray⟦1;;2⟧]
```

Out[∘]//TableForm=

$$\lambda[x]^4 y[0][x] + \rho[x] \times y[0][x] == 0$$

$$6\lambda[x]^2 y[0][x]\,\lambda'[x] + 4\lambda[x]^3 y[0]'[x] == 0$$

Поиск функции $\lambda(x)$

In[∘]:=
```
Eq0 = EqArray⟦1⟧ /. {y[0][x] → 1}
```

Out[∘]=

$$\lambda[x]^4 + \rho[x] == 0$$

In[∘]:=
```
SolLambda = Flatten[Solve[Eq0, λ[x]]]
```

Out[∘]=

$$\left\{\lambda[x] \to -(-1)^{1/4}\rho[x]^{1/4},\ \lambda[x] \to (-1)^{1/4}\rho[x]^{1/4},\ \lambda[x] \to -(-1)^{3/4}\rho[x]^{1/4},\ \lambda[x] \to (-1)^{3/4}\rho[x]^{1/4}\right\}$$

Нулевое приближение

In[∘]:=
```
Eq1 = Simplify[EqArray⟦2⟧ /. {SolLambda⟦#⟧, λ'[x] → D[SolLambda⟦#, 2⟧, x]}, ρ[x] > 0] &;
Eq1[1]
```

Out[∘]=

$$3\, y[0][x]\, \rho'[x] + 8\, \rho[x]\, y[0]'[x] == 0$$

In[∘]:=
```
Sol0 = Flatten[Table[DSolve[Eq1[i], y[0][x], x]⟦1⟧, {i, 4}]]
```

Out[∘]=

$$\left\{y[0][x] \to \frac{\mathbb{C}_1}{\rho[x]^{3/8}},\ y[0][x] \to \frac{\mathbb{C}_1}{\rho[x]^{3/8}},\ y[0][x] \to \frac{\mathbb{C}_1}{\rho[x]^{3/8}},\ y[0][x] \to \frac{\mathbb{C}_1}{\rho[x]^{3/8}}\right\}$$

Ответ

In[ ]:=
```
Sol = Table[y[x] → (Y /. Sol0⟦i⟧) /. {Λ[x] → (Integrate[λ[x], {x, 0, x}] /. SolLambda⟦i⟧)}, {i, 4}]
```

Out[ ]=

$$\left\{y[x] \to \frac{e^{\frac{\int_0^x -(-1)^{1/4}\rho[x]^{1/4}\,dx}{\mu}}\,C_1}{\rho[x]^{3/8}},\ y[x] \to \frac{e^{\frac{\int_0^x (-1)^{1/4}\rho[x]^{1/4}\,dx}{\mu}}\,C_1}{\rho[x]^{3/8}},\ y[x] \to \frac{e^{\frac{\int_0^x -(-1)^{3/4}\rho[x]^{1/4}\,dx}{\mu}}\,C_1}{\rho[x]^{3/8}},\ y[x] \to \frac{e^{\frac{\int_0^x (-1)^{3/4}\rho[x]^{1/4}\,dx}{\mu}}\,C_1}{\rho[x]^{3/8}}\right\}$$